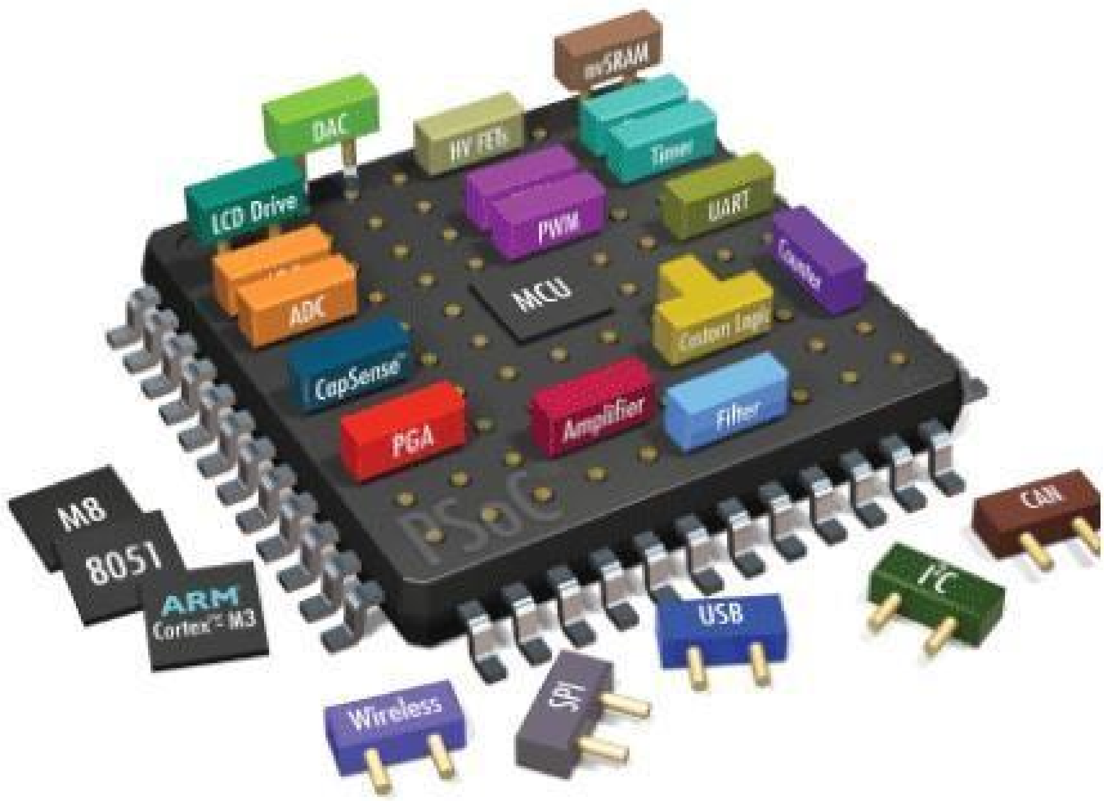
HITEC UNIVERSITY, TAXILA

DEPARTMENT OF COMPUTER ENGINEERING

Lab # 02: Introduction to MTS-51 Trainer board Inside's with demo examples

Embedded Systems

Introduction to MTS-51 Trainer board Inside's with demo examples



Dated:

Week 02

Semester:

Fall 2023

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MTS-51 8051 MICROCOMPUTER TRAINER

FEATURES & BENEFITS

1. With the ISP and IAP functions of Philips's P89C51RD+ / P89C51RD2 control chip, the program codes are able to download to flash memory through series port interface, showing the results at real time
2. Built-in power supply Input AC power: 110/ 220V, 50/60Hz Output DC power: 12V/3A, 5V/3A
3. Reserving external connect pins for advanced experiments
4. Plenty of experiments to do various basic I/O control applications
5. The single chip of the trainer can be replaced by INTEL's 8751/52 series (without ISP function) and ATMEL's AT89C51/52 series (with ISP function)

6. MTS-51 trainer is a training equipment for learning basic control applications with 8051 microcontrollers. With various common I/O components and experiments, users are able to learn the 8051-control implementation efficiently

SPECIFICATION

1. P89C51 single chip x 1, with ISP (In System Programming) function
2. LCM x 1, back light (20 x 2 lines) LCD module
3. 7-segment display x 4
4. LED-bar 10-bit x 2
5. 4 x 4 matrix keyboard x 1
6. 12V stepping motor x 1, 200 steps, with A, B, A, B coil output connector
7. Photochopper x 2 PH1 for interrupt request signal and interdict counter PH2 for counter, pulse counter or counter control and interdict counter
8. IC555 x 1, instable oscillating circuit for pulse signal output
9. 8 x 8 dot matrix LED x 1
10. RS-232 interface x 1, interface for ISP function
11. 8-bit DIP switch x 3, for circuit start control
12. Micro speaker x 1
13. 10 x 2 extend socket x 1, for P0 and P2 output
14. Through RxD, TxD connector for multi-chip transfer control

EXAMPLES

1. LED display control
2. Dot Matrix LED Control
3. Step Motor Control
4. Input Port Expansion
5. Pulse Counter
6. Speaker Control
7. Serial Communications
8. 7-Segment Display Control
9. Matrix Keyboard Control
10. Output Port Expansion

11. Photo Interrupter Control
12. Timer/Counter
13. LCM Display Control

General Instructions:

1. Turn off the power switch
2. Insert the power cable
3. Turn on the power switch
4. Place DIP switch SW1 no. 5 to ON position
5. Welcome screen is displayed on the LCM

Lab Tasks:

1. Show the seven-segment display 3 and 4 by pressing 1. Write down the steps to have a successful test. Take the snapshot of the final output generated by test.
2. Show the led bar 2 by pressing 2. Write down the steps to have a successful test. Take the snapshot of the final output generated by test.
3. Show the stepping motor by pressing 3. Write down the steps to have a successful test. Take the snapshot of the final output generated by test.
4. Show the IC 555 timer to speaker by pressing 4. Write down the steps to have a successful test. Take the snapshot of the final output generated by test.
5. Show the IC 555 timer to seven segment display by pressing 5. Write down the steps to have a successful test. Take the snapshot of the final output generated by test.
6. Show the 8x8 dot matrix by pressing 6. Write down the steps to have a successful test. Take the snapshot of the final output generated by test.
7. Show the seven-segment display 1 and 2 by pressing 7. Write down the steps to have a successful test. Take the snapshot of the final output generated by test.
8. Show the led bar 1 by pressing 8. Write down the steps to have a successful test. Take the snapshot of the final output generated by test.
9. Show the photo interrupter (PH1) by pressing 9. Write down the steps to have a successful test. Take the snapshot of the final output generated by test.
10. Show the photo interrupter (PH2) by pressing 10. Write down the steps to have a successful test. Take the snapshot of the final output generated by test.
11. Show the parallel in serial out shift register by pressing 11. Write down the steps to have a successful test. Take the snapshot of the final output generated by test.

Lab Performance Evaluation:
Read and practice the manual. Performance is evaluated at the end of lab based on successfully running and understanding of examples, tasks and viva using defined rubrics.

Lab Report Evaluation:
Submit the Lab Report by writing all the examples and tasks which must include the following:

1. Brief description (3-5 lines)
2. The steps
3. The results (Screenshot)

Important note: Lab Report must be according to the Lab Report Format available on Google Classroom and must be submitted on time. Two similar reports will get zero marks. So, don't try to copy your fellow reports. Lab report must be submitted in group of three maximum.